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Project 8

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Introduction

This report describes the design and implementation of a tool that serves a virtual flat-file view over data stored in a relational representation. The central idea is to preserve the queryability and compact storage of a tabular backend while exposing the data through a conventional file interface that can be inspected with standard filesystem operations. In this work, compressed storage is provided through Apache Parquet, a columnar format designed for efficient storage and retrieval with built-in compression and encoding support, while DuckDB is used as the embedded analytical database managing the relational view of the data. The file-oriented presentation layer is implemented with FUSE, which enables user-space programs to export filesystems to the kernel and thereby present generated content as ordinary files. :contentReferenceoaicite:0

As a concrete demonstration, the tool is applied to genomic variant data represented in Variant Call Format (VCF). VCF is a text-based format consisting of meta-information lines, a header, and tab-delimited variant records with genotype information for one or more samples, making it a natural target for virtual flat-file materialization from relational tables. This setting provides a useful test case because VCF combines structured schema-like fields with a strict textual serialization, allowing the system to illustrate how relational records can be reconstructed into a standards-compliant, human-readable file without duplicating the underlying dataset. :contentReferenceoaicite:1

Formatting

This document use Markdown and you can look at this tutorial.

Subsection level 2

Please keep sections to a maximum of only two levels.

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Citation Typing Ontology annotation

You can use CiTO annotations, as explained in this BioHackathon Europe 2021 write up and this CiTO Pilot. Using this template, you can cite an article and indicate *why* you cite that article, for instance DisGeNET-RDF (Queralt-Rosinach et al., 2016).

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- providesDataFor
- obtainsSupportFrom
- discusses
- extends
- agreesWith
- disagreesWith
- updates
- citation: generic citation

Results

Discussion

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Acknowledgements

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References

Queralt-Rosinach, N., Pinero, J., Bravo, À., Sanz, F., & Furlong, L. I. (2016). DisGeNET-RDF: harnessing the innovative power of the Semantic Web to explore the genetic basis of diseases. *Bioinformatics*, *32*(14), 2236–2238. https://doi.org/10.1093/bioinformatics/btw214 **[cito:citesAsAuthority]**

Author contributions

Elias Crum: VCF to TSV conversion, duckDB data ingestion, manuscript writing + editing
Benno Kruit: FUSE virtual file generation from duckDB data, manuscript writing + editing